



Operating System - 21CS2404

Branch: CSE / CE

Assignment No. 1

Assigned Date: 04/02/26

Due Date: 11/02/26

Sr. No.	Questions :- Module 1, 2
1	Define operating system. Write down Services / functions / tasks of operating system.
2	Explain different types of Operating Systems.
3	Explain similarities and dissimilarities between thread and process.
4	What is process scheduling? Explain all types of process schedulers.
5	What is Process? Draw Five State Process Model and Explain it.
6	What is thread? Explain advantages of thread. Explain any one type of thread in details.
7	What is PCB? Discuss its major fields.
8	Explain Process creation and Process termination in detail.
9	Discuss the objectives of CPU Scheduling and explain in detail about CPU scheduling criteria.
10	Explain Context Switching and its impact on System Performance.
11	Define Response Time and Turn Around time.

12	Consider the following set of processes, with the length of the CPU-burst time given in milliseconds: Calculate average Turnaround time and average waiting time of following processes using Round robin algorithm. Take quantum time=4ms and context switch overhead=1ms. <table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>2</td><td>10</td></tr><tr><td>P2</td><td>5</td><td>8</td></tr><tr><td>P3</td><td>0</td><td>5</td></tr><tr><td>P4</td><td>3</td><td>6</td></tr><tr><td>P5</td><td>1</td><td>9</td></tr></table>	Process	Arrival Time	Burst Time	P1	2	10	P2	5	8	P3	0	5	P4	3	6	P5	1	9
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13	Differentiate between preemptive and non-preemptive scheduling. Solve following by Shortest Remaining Time First- SRTF (preemptive) and Shortest Job First (non-preemptive). Draw Gantt chart, Average Waiting Time and Average Turnaround Time. Which one is better as per average turnaround time? <table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>0</td><td>6</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>5</td></tr><tr><td>P4</td><td>4</td><td>9</td></tr><tr><td>P5</td><td>6</td><td>8</td></tr></table>	Process	Arrival Time	Burst Time	P1	0	6	P2	1	4	P3	2	5	P4	4	9	P5	6	8
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